

Presentation Speech (~6 minutes)

Each section corresponds to one slide. Advance when you reach the next [SLIDE] marker.

[SLIDE 1] Title

Title slide is displayed as you walk up. No speaking — wait for audience to settle.

[SLIDE 2] Thesis

Living things are not machines.

That is the thesis. That is the finding. That is what seven layers of quantitative evidence confirm.

My name is Sage Clokey, and my capstone asks a question that sounds simple but has enormous consequences for how we do molecular biology: does life operate like a centrally planned economy, or like a free market?

[SLIDE 3] The Central Planner's Promise

Modern synthetic biology assumes the answer is central planning. Pick the right promoter. Optimize the codons. Calculate the expression ratios. Assemble the parts. The engineer is the planner, the genes are the workers, and the construct is the five-year plan. And it fails — at roughly the same rate and for roughly the same reasons that central planning fails in human economies. Kwok documented in *Nature* that most designed genetic circuits do not work as predicted on the first attempt. Purnick and Weiss showed that complexity hits a wall beyond a handful of components. The standard explanation is that biology is “complex” and our models are “incomplete.”

This paper argues the standard explanation is wrong. The problem is not insufficient data. The problem is that the knowledge required to coordinate a living system exists only in distributed form — scattered across thousands of molecular agents, each responding to local conditions that no external observer can fully access. This is not a limitation of current technology. It is a structural feature of life.

[SLIDE 4] The Knowledge Problem — From Vienna to the Genome

The insight comes from the Austrian school of economics. Hayek showed in 1945 that the knowledge required for economic coordination never exists in concentrated form — it exists only as dispersed bits held by separate individuals. No central planner can gather it. Mises proved in 1920 that rational calculation is impossible without prices. Menger showed that complex institutions arise spontaneously from individual action. Every principle they discovered about human cooperation, life discovered first — and wrote it in DNA.

So I tested it. Seven layers of analysis.

[SLIDE 5] Layer 1 — Network Topology

Layer 1: Network topology. I built five biological networks from public databases — the E. coli gene regulatory network, protein-protein interaction networks, metabolic networks — and compared them against five synthetic architectures including a star graph, which is the topology of central planning. The result: there is no master node. Biological networks survive removing 37% of their most connected nodes before fragmenting. The star graph collapses at 1.9%. That is a 19-to-1 robustness ratio. The structural cost of centralization, measured in network connectivity.

[SLIDE 6] Layer 1b — Single-Cell Economy

Layer 1b: Single-cell economics. I analyzed 2,638 human immune cells across 8 cell types. Each cell type specializes — division of labor without anyone assigning roles. The communication network has a betweenness Gini of zero — perfectly distributed, no gatekeeper. Remove any single cell type and 75% of communication survives. This is Menger's spontaneous order made visible.

[SLIDE 7] Layer 2 — Metabolic Simulation

Layer 2: Metabolic simulation. I built an agent-based economy with 13 metabolic pathway agents. Under stable conditions, the central planner achieves higher output — it can see the whole board. But when I removed an agent — a structural perturbation — the distributed system retained 71% of GDP through local self-correction. The centralized system retained only 53%. An 18-point advantage for the market under stress. The planner's fixed allocation becomes wrong the moment conditions change, and it has no mechanism to discover the new optimum.

[SLIDE 8] Layer 2b — FBA

Layer 2b: Flux balance analysis. I gave the planner perfect information — the iML1515 genome-scale model of *E. coli*, 2,712 reactions, 1,877 metabolites, 1,516 genes. This is the omniscient planner. The Gosplan with a supercomputer. It achieves 70% accuracy on gene knockout predictions. The 30% failure is not noise — it is structural. The failures are genes essential for regulatory reasons the linear program cannot encode: allosteric feedback, protein folding, expression timing. Knowledge that exists only in the local state of each molecular agent. And here is the irony — the LP must compute shadow prices to solve its optimization. Prices. The very thing Hayek said the market produces for free.

[SLIDE 9] Layer 3 — Cross-Species Trade

Layer 3: Cross-species trade. Gene transfer between species follows trade network rules. Costs scale with evolutionary distance — human to axolotl is cheap, prokaryote to eukaryote is expensive. Forced codon optimization destroys information encoded in rare codons. Codon harmonization preserves it. And trade blocs emerge spontaneously from shared evolutionary history — nobody designed them.

[SLIDE 10] Layer 4 — Immune System

Layer 4: The immune system. This is where the data gets striking. Textbooks call V(D)J recombination and somatic hypermutation “random.” They are not. AID targets hotspot motifs at 19-to-1 over coldspots. V segment usage is biased 10 to 20x toward preferred segments — and the same bias appears in unrelated individuals, Spearman rho approaching 1.0. And public clonotypes — identical T cell receptor sequences appearing in unrelated people at rates exceeding random expectation by 10-to-the-15th-fold. Independent immune systems arriving at the same molecular solution without communication. That is not randomness. That is convergent distributed discovery.

[SLIDE 11] Layer 5 — Whole Genome

Layer 5: The whole genome. CpG mutations occur at 15 to 40x baseline from 1% of sequence contexts. Twenty percent of genes show tissue specificity above tau 0.95 — expressed almost exclusively in one tissue. And 35 convergent evolution events across 17 traits reproduce the same amino acid substitutions in lineages separated by up to

1.5 billion years. The same molecular solutions, found independently, across kingdoms and eons.

[SLIDE 12] Evidence Summary Table

Seven layers. Same answer. The knowledge is distributed.

Pause briefly. Let the audience read the table.

[SLIDE 13] Six Design Principles

What does this mean for the practicing molecular biologist? Six design principles. Read the economy before entering it — use FBA shadow prices. Build feedback loops, not fixed rates. Distribute control across the pathway. Harmonize codons instead of force-optimizing them. Let the system evolve. And design consortia, not monoliths.

[SLIDE 14] Closing

The cell is not a chassis waiting to be programmed. It is a running economy — 4,400 genes coordinating through distributed feedback. Every gene the engineer inserts is a new firm entering an existing market. The planner ignores the existing order and tries to override it. The sagent reads the economy, reduces trade barriers, grows feedback loops, and lets the system discover what no planner could predict.

The garden is still growing. The question is not how to program life. It is how to join it.

[SLIDE 15] Thank You

Thank you.

Take questions.

Approximate word count: ~880 words / ~6 minutes at moderate speaking pace